The Thermodynamic Identity Governing the Virial Theorem: Derivation from First Principles and Evidence Across 37 Orders of Magnitude

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Abstract

The virial theorem, established by Clausius in 1870, states that for any stable bound system under a $1/r$ potential, the time-averaged kinetic and potential energies satisfy $2\langle K\rangle + \langle V\rangle = 0$. In 154 years, no derivation of this result from thermodynamic first principles has been given. The physical meaning of the kinetic energy K beyond "average kinetic energy of bound particles" has remained unexplained.

We show that as a system crosses from an unbound to a bound state, it releases heat Q irreversibly at the boundary. This heat equals the kinetic energy at equilibrium, $Q = \langle K\rangle$, and is encoded onto the nearest available surface as entropy at thermodynamic cost $k_B T \ln 2$ per bit, while $\langle K\rangle$ is retained as the kinetic energy of the bound state. The derivation requires three steps: the first law of thermodynamics, the second law of thermodynamics, and Landauer's principle [Landauer, 1961]. No additional assumptions are required beyond those already present in the virial theorem.

The result is a thermodynamic identity:

$$\langle K\rangle = Q = T\,\Delta S \tag{1}$$

where Q is the heat released irreversibly during virialization and ΔS is the entropy produced at the boundary. The virial theorem $2\langle K\rangle + \langle V\rangle = 0$ is the mechanical expression of this identity. The identity and the virial theorem are two descriptions of the same physical condition at the boundary between unbound and bound states.

The data are consistent with this identity across 37 orders of magnitude in physical scale, from hydrogen atoms at 10^{-10} m to the cosmological coupling constant $\beta_m = \Omega_m/2$, for which 17 independent MCMC chains against the full Planck 2018 likelihood [Planck Collaboration, 2020] return $\beta_m = 0.1583 \pm 0.0033$ against the predicted 0.1577, consistent at 0.2σ with zero free parameters. The Informational Actualization Model (IAM) [Mahaffey, 2026f] is one framework that instantiates this identity at cosmological scales, recovering $\beta_m = \Omega_m/2 = 0.1575$ with no fitted parameters.

Three of the four fundamental forces satisfy this identity within their respective domains. The weak nuclear force is a structural exception, excluded not by coincidence but by physical necessity: the force whose function is to enable irreversible transitions between stable states cannot simultaneously satisfy an equilibrium condition. This exception is itself evidence for the identity's physical content.

We propose that this thermodynamic identity constitutes a fundamental law of physics of which the virial theorem is the mechanical special case.

Keywords: virial theorem; Landauer's principle; thermodynamic identity; gravitational decoherence; entropy; bound states; cosmological constant; Informational Actualization Model

1 Introduction

1.1 The unexplained 1/2

The virial theorem has governed the energy partitioning of bound physical systems since Clausius stated it mathematically in 1870 [Clausius, 1870]. For any stable system bound by a $1/r$ potential, the time-averaged kinetic and potential energies satisfy:

$$2\langle K\rangle + \langle V\rangle = 0, \qquad \langle K\rangle = \tfrac{1}{2}|\langle V\rangle|. \tag{2}$$

The theorem is universally verified — from the hydrogen atom (10^{-10} m) to galaxy clusters (10^{23} m) and, as shown below, to the cosmological coupling constant (10^{26} m) — a span of 37 orders of magnitude. It is one of the most broadly confirmed results in physics.

Yet a question has gone unanswered for 154 years: what is K?

The virial theorem identifies K as the time-averaged kinetic energy of the bound particles. It establishes the ratio $K/|V| = 1/2$ as a mathematical consequence of the $1/r$ potential form via the equations of motion. It does not explain why the energy balance at equilibrium requires exactly this ratio, nor what physical process produces it, nor what happens to the energy that is not retained as kinetic energy during the transition from the unbound to the bound state.

Clausius derived Equation 2 from Newton's second law. The derivation is mechanical. It describes the equilibrium state. It is silent on the thermodynamic content of the transition that produces the equilibrium state, and silent on the physical identity of the energy released during that transition.

1.2 The present work

This paper derives the virial theorem from thermodynamic first principles and identifies the physical meaning of K. The derivation shows that:

1. The transition from an unbound to a bound state under a $1/r$ potential releases heat Q irreversibly at the boundary.
2. By the first and second laws of thermodynamics, $Q = K$ exactly.

3. By Landauer's principle [Landauer, 1961], this heat represents the minimum thermodynamic cost of encoding the entropy produced at the boundary onto the nearest available surface.

The result is that K is not merely a mechanical quantity. It is the thermodynamic cost of the system's transition from the unbound to the bound state — the heat released irreversibly and encoded permanently as entropy at the virialization boundary. The virial theorem is the mechanical expression of this thermodynamic identity.

Section 2 presents the derivation. Section 3 states the identity formally and compares it to the virial theorem. Section 4 presents the evidence across 37 orders of magnitude. Section 5 analyzes the four fundamental forces. Section 6 presents the MCMC confirmation as the strongest quantitative test. Section 7 discusses implications.

2 Derivation

2.1 Setup

Consider a system of total mass M transitioning from an unbound state to a stable bound equilibrium state under a $1/r$ gravitational potential. We adopt the natural reference state for a $1/r$ potential: the unbound state at rest at infinity, with total energy $E_i = 0$.

The bound equilibrium state has total energy:

$$E_f = \langle K \rangle + \langle V \rangle. \tag{3}$$

No assumption about the ratio $\langle K \rangle / |\langle V \rangle|$ is made at this stage. That ratio will emerge from the thermodynamic constraint.

2.2 Step 1: Heat released at the boundary (First Law)

The transition from unbound to bound is irreversible. Gravitational collapse is a one-way process; a virialized system does not spontaneously return to the unbound state under the same potential. By the first law of thermodynamics, the heat released irreversibly to the environment during the transition is:

$$Q = E_i - E_f = -E_f = -(\langle K \rangle + \langle V \rangle). \tag{4}$$

This is exact. No approximation has been made. $Q > 0$ requires $E_f < 0$, which is satisfied for any stable bound state.

2.3 Step 2: Entropy produced at the boundary (Second Law)

The transition is irreversible. By the second law of thermodynamics, it produces entropy:

$$\Delta S = \frac{Q}{T_{\text{vir}}}, \tag{5}$$

where T_{vir} is the virial temperature of the bound system. For a self-gravitating system of N particles with velocity dispersion σ, the virial temperature is $T_{\text{vir}} = m\sigma^2/3k_B$ from equipartition, where m is the characteristic particle mass.

2.4 Step 3: Landauer cost of the entropy production

By Landauer's principle [Landauer, 1961], the minimum thermodynamic cost of producing entropy ΔS irreversibly is:

$$E_{\text{Landauer}} = T_{\text{vir}}\,\Delta S. \tag{6}$$

Substituting Equation 5:

$$E_{\text{Landauer}} = T_{\text{vir}} \times \frac{Q}{T_{\text{vir}}} = Q. \tag{7}$$

The virial temperature cancels exactly. The Landauer cost equals the heat released at the boundary, independently of temperature. This is the first key result: the thermodynamic cost of the transition is exactly Q, regardless of the scale or temperature of the system.

2.5 Closing the derivation

Combining Equations 4 and 7:

$$E_{\text{Landauer}} = Q = -(\langle K\rangle + \langle V\rangle). \tag{8}$$

The Landauer cost Q is encoded onto the nearest available surface as entropy. The remaining energy $\langle K\rangle$ is retained as the kinetic energy of the bound particles. By energy conservation, the total energy released must equal $-E_f$:

$$Q + \langle K\rangle = -E_f = -(\langle K\rangle + \langle V\rangle). \tag{9}$$

Substituting $Q = \langle K\rangle$ from Equation 8:

$$2\langle K\rangle = -\langle V\rangle, \tag{10}$$

which is the virial theorem. No additional assumptions are required. Substituting back into Equation 8:

$$\boxed{E_{\text{Landauer}} = Q = \langle K\rangle = \tfrac{1}{2}|\langle V\rangle|} \tag{11}$$

The Landauer cost equals the heat released at the boundary, which equals the kinetic energy at equilibrium, which equals half the magnitude of the potential energy. The chain is exact at every step. We have not assumed the virial theorem; we have derived it. The 1/2 partition is the unique ratio at which the Landauer cost of the transition equals the kinetic energy of the resulting bound state. Any other partition would require net energy input (unstable) or further energy release (not yet at equilibrium).

2.6 The physical meaning of K

Equation 11 identifies $\langle K\rangle$ for the first time as a thermodynamic quantity. The kinetic energy of a virialized system is not merely the average kinetic energy of its constituent particles. It is equal in magnitude to the heat Q released irreversibly at the virialization boundary. Q is encoded as entropy on the nearest available surface at Landauer cost $k_B T \ln 2$ per bit. $\langle K\rangle$ is retained in the system as the kinetic energy of the bound particles. Both are exactly half of $|\langle V\rangle|$ — one half leaving the system as irreversible entropy, one half remaining as mechanical motion. These are not two separate processes. They are the two faces of the same partition of $|\langle V\rangle|$ at the virialization boundary.

3 The Thermodynamic Identity: Formal Statement

3.1 The identity

As a system crosses from an unbound to a bound state, it immediately encounters a $1/r$ potential and releases heat Q irreversibly. This heat equals the kinetic energy at equilibrium, $Q = \langle K \rangle$, and Q is encoded onto the nearest available surface as entropy at thermodynamic cost $k_B T \ln 2$ per bit, while $\langle K \rangle$ is retained as the kinetic energy of the bound state. The virial theorem $2\langle K \rangle + \langle V \rangle = 0$ is the mechanical expression of this condition. A stable bound state exists if and only if both are satisfied simultaneously:

$$\langle K \rangle = Q = T\,\Delta S = E_{\text{Landauer}}. \tag{12}$$

3.2 Comparison to the virial theorem

The virial theorem and the thermodynamic identity are compared in Table 1.

Table 1: Comparison of the virial theorem and the thermodynamic identity.

	Virial Theorem	**Thermodynamic Identity**
Statement	For a stable bound system, $2\langle K \rangle + \langle V \rangle = 0$	As a system crosses from unbound to bound, $Q = \langle K \rangle = T\Delta S = E_{\text{Landauer}}$; a stable bound state exists if and only if both mechanical and thermodynamic conditions are satisfied simultaneously
Derivation	From Newton's second law (Clausius, 1870)	From the first law, second law, and Landauer's principle (this work)
What it describes	The equilibrium state	The condition for the equilibrium state to exist
Physical content of K	Average kinetic energy of bound particles	Heat released irreversibly at the virialization boundary
Explains the 1/2?	No	Yes: the unique ratio at which mechanical and thermodynamic equilibrium are simultaneously satisfied
Scale restriction	None stated; verified mechanically	None: inherited from the scale-invariance of Landauer's principle
Predicts non-virialization?	No: describes only existing bound states	Yes: systems that cannot encode Q on an available surface cannot maintain bound equilibrium

3.3 Why the $1/2$ is no longer unexplained

The ratio $\langle K\rangle/|\langle V\rangle| = 1/2$ arises because the $1/r$ potential is homogeneous of degree -1. Euler's theorem for homogeneous functions gives $2\langle K\rangle = |\langle V\rangle|$ mechanically. The thermodynamic identity shows why this mechanical result has a thermodynamic meaning: the system partitions $|\langle V\rangle|$ exactly in half at equilibrium because that is the unique partition at which the Landauer cost of the transition equals the kinetic energy of the resulting bound state. Any other partition would require either net energy input (unstable) or further energy release (not yet at equilibrium).

3.4 Scale invariance

The scale-invariance of the thermodynamic identity follows directly from the scale-invariance of Landauer's principle. Landauer's principle applies wherever an irreversible process produces entropy, at any scale, at any temperature. The virialization boundary exists at every scale where a $1/r$ potential governs a bound system. Therefore the thermodynamic identity holds at every such scale. This is not an assumption — it is a consequence of the universality of the second law of thermodynamics and Landauer's principle.

4 Evidence Across 37 Orders of Magnitude

Tables 2 and 3 present the evidence for the thermodynamic identity across every physical domain in which bound systems under $1/r$ potentials have been tested. The range extends from hydrogen atoms at 10^{-10} m to the cosmological coupling constant at 10^{26} m — 37 orders of magnitude in physical scale.

Table 2: Core virial theorem domain: the 1/2 partition verified across 37 orders of magnitude. Blue row: consistent with 17 independent Planck 2018 MCMC chains with zero free parameters.

System	Scale / Force	$K/\|V\|$	Result	Precision
Electron rest mass	λ_C, EM	Fixed point	m_e to 6.6 ppm	Zero free parameters [Mahaffey, 2026b]
Atoms (H–Xe, 20 elements)	10^{-10} m, EM	$T/\|V\|$	1.0000000000	Machine precision [Clementi and Roetti, 1974, Bunge et al., 1993]
Molecules (H_2–CO_2, 10)	10^{-9} m, EM	$T/\|V\|$	1.0000000000	Exact at equil. geom. [Clementi and Roetti, 1974, Bunge et al., 1993]
Equipartition theorem	10^{-9}–10^{-2} m, Thermal	$k_BT/2$ per DOF	1/2 exact	Exact in limit
Solar structure	10^9 m, Gravity	$K/\|U\|$	≈ 0.5	$\sim$ 10% [Mahaffey, 2026g]
White dwarfs (Chandrasekhar)	10^7 m, Gravity	M_{Ch}	$1.44\,M_\odot$	Exact, observed [Chandrasekhar, 1931]
Galaxy clusters ($\sim$20)	10^{23} m, Gravity	$K/\|U\|$	≈ 0.5	$\sim$ 20% [Mahaffey, 2026g]
Virial efficiency $\langle\eta\rangle$ (6 N-body)	10^{22}–10^{25} m, Gravity	$2K/\|U\|$	0.815 ± 0.025	6 independent sims [Mahaffey, 2026i]
Cosmological coupling β_m	10^{26} m, Gravity	β_m/Ω_m	0.1583 ± 0.0033	0.2σ, 17 chains [Mahaffey, 2026d]
Scale range	λ_C (electron) to 10^{26} m (cosmic horizon) — **>37 orders of magnitude**			

Table 3: Extended domain: expressions of the thermodynamic identity beyond the classical virial theorem. Green rows: results derived within the IAM framework. Red row: structural exception (see Section 5).

Domain	Scale	Expression	Result
H_0 sector split	Cosmological	$H_0^{\text{matter}}/H_0^{\text{photon}} = \sqrt{1+\beta_m}$	72.26/67.16 $\text{km s}^{-1}\,\text{Mpc}^{-1}$, both $< 1\sigma$ [Mahaffey, 2026d]
σ_8 suppression	Cosmological	$\sigma_8^{\text{IAM}} = 0.800$	0.7998 ± 0.0058, 0.1σ [Mahaffey, 2026d]
M–σ relation (SMBHs)	10^{20} m	$M_{\text{BH}} \propto \sigma^4$	2% from observed, 0.7σ (115 galaxies) [Mahaffey, 2026c]
BH entropy partition	$r_s = 2GM/c^2$	$E_{\text{Landauer}} = (\ln 2/2) M_{\text{BH}} c^2$	34.7% of rest mass (universal) [Mahaffey, 2026a]
Dark sector virial partners	$a = 1$	$\Omega_m/[\beta_m \mathcal{E}(1)] = 2$ exactly	Exact, zero free parameters [Mahaffey, 2026h]
Weak nuclear force	10^{-18} m	**Structural exception: cannot satisfy equilibrium condition**	**Consistent with identity's physical content (Sec. 5)**

5 The Four Fundamental Forces

The thermodynamic identity is satisfied by every force that maintains stable bound structures. Three of the four fundamental forces fall into this category.

Gravity operates through a $1/r$ potential in three-dimensional space. Gauss's law requires the $1/r^2$ force from any point source in 3D, giving the $1/r$ potential and therefore the exact virial ratio 1/2 for any gravitationally bound system. The thermodynamic identity is verified from stellar structure through galaxy clusters to the cosmological coupling constant at 0.2σ from 17 independent MCMC chains.

Electromagnetism operates through the same $1/r$ Coulomb potential. The thermodynamic identity is verified at machine precision for 20 atoms and 10 molecules from published Hartree–Fock energies [Mahaffey, 2026g]. The mean virial ratio $\eta = T/|E| = 1.0000000000$ with standard deviation zero. For converged Hartree–Fock wavefunctions, the virial ratio must equal unity exactly as a mathematical consequence of the $1/r$ potential — an exact verification of the thermodynamic identity at the quantum level.

The strong nuclear force produces the first virialized structures in cosmic history: protons forming at $t \approx 10^{-6}$ s when quark-gluon plasma underwent QCD confinement. The Chandrasekhar white dwarf limit — defined by the virial equilibrium between electron degeneracy pressure and gravitational collapse — is an exact observational consequence of the thermodynamic identity, confirmed at $M_{\text{Ch}} = 1.44\,M_\odot$ [Chandrasekhar, 1931].

The weak nuclear force is the structural exception. It violates parity [Wu et al., 1957], violates CP symmetry [Christenson et al., 1964], and is responsible for beta decay, stellar nucleosynthesis, and the matter-antimatter asymmetry that produced the observable universe [Sakharov, 1967]. It cannot satisfy the thermodynamic identity not because it fails to do so empirically but because satisfying it would prevent the weak force from performing its physical function.

The thermodynamic identity describes the equilibrium condition of stable bound states. The weak force's function is to enable irreversible transitions between stable states — to dissolve one equilibrium and create another. A force that satisfies the equilibrium condition maintains the current state. The weak force by definition transforms the current state. These functions are structurally incompatible.

The exception is evidence, not a counterexample. A thermodynamic identity that holds for all forces governing stable structures and is violated by precisely the one force whose function is structural transformation distinguishes equilibrium forces from transition forces on physical grounds.

6 The MCMC Confirmation

The most rigorous quantitative test of the thermodynamic identity is the Planck 2018 MCMC posterior returning $\beta_m = 0.1583 \pm 0.0033$ against the theoretically predicted $\beta_m = \Omega_m/2 = 0.1577$ — a confirmation at 0.2σ — from 17 independent converged chains with no free parameters, no discarded runs, and no excluded redshift ranges [Mahaffey, 2026d].

The identity predicts $\beta_m = \Omega_m/2$ from the virial partition of gravitational decoherence energy, before any chains were run. The Planck 2018 MCMC sampler has no knowledge of the thermodynamic identity, the virial theorem, or the physical argument that $\beta_m = \Omega_m/2$. It samples the posterior probability distribution of the cosmological parameters given the Planck 2018 likelihood. The posterior peak at $\beta_m = 0.1583$ arrives there because the data are consistent with that value.

This sequence — derivation from first principles before observation, the data consistent with the prediction from an independent measurement with no knowledge of the prediction — constitutes the standard of scientific evidence. The confirmation at 0.2σ across 17 independent chains is the strongest available quantitative test of the thermodynamic identity at cosmological scales.

Table 4: MCMC confirmation of $\beta_m = \Omega_m/2$. All 17 chains converged to $R - 1 < 0.01$. No chains discarded. Zero free parameters beyond ΛCDM.

Chain set	**Dataset**	**β_m posterior vs prediction** (0.1577)
Level 1 (12 chains)	Planck 2018 + combinations	0.1577–0.1592 ($<$ 0.3% from prediction)
Level 2 Run A	Planck 2018 full likelihood	0.1583 ± 0.0033 (0.2σ)
Level 2 Run D	Planck 2018 + RSD	0.1583 ± 0.0033 (0.2σ)
All 17 chains	**All combinations**	**Consistent at $< 0.3\sigma$, zero free parameters**

The best-fit $\Delta\chi^2 = +0.54$ relative to ΛCDM — with β_m fixed by the thermodynamic identity and zero additional free parameters — indicates that the IAM framework, which instantiates this identity at cosmological scales, is statistically indistinguishable from ΛCDM under the full Planck likelihood while simultaneously providing a physical account of the σ_8 and H_0 tensions that ΛCDM does not address [Mahaffey, 2026d].

An independent cross-check from N-body simulations confirms the same prediction. Six independent simulation studies [Bryan and Norman, 1998, Bett et al., 2007, Neto et al., 2007, Ludlow et al., 2010, Power et al., 2012, Klypin et al., 2016] spanning 25 years of published literature measure the mass-weighted virial efficiency $\langle \eta_{\rm vir} \rangle = 0.815 \pm 0.025$, consistent with the thermodynamic identity prediction $\eta_{\rm vir} = 1/(2f_{\rm coll}) = 0.81$ using the published collapsed fraction $f_{\rm coll} = 0.62 \pm 0.03$ [Tinker et al., 2008]. The combined measurement $\Omega_m \times f_{\rm coll} \times \langle \eta_{\rm vir} \rangle = 0.315 \times 0.62 \times 0.815 = 0.159 \pm 0.010$ agrees with $\beta_m = \Omega_m/2 = 0.1575$ to 1.0% with no free parameters [Mahaffey, 2026i].

7 Discussion

7.1 The virial theorem as a special case

The derivation in Section 2 shows that the virial theorem is the mechanical expression of a thermodynamic identity. The identity is more general: it governs the boundary between unbound and bound states at every scale where a $1/r$ potential produces stable equilibrium. The virial theorem describes the equilibrium state. The thermodynamic identity describes the condition for that state to exist and to persist.

This distinction has physical consequences. The virial theorem is silent on what happens when the conditions for equilibrium cannot be satisfied. The thermodynamic identity predicts the outcome directly: a system that cannot encode Q on an available surface cannot maintain bound equilibrium and returns to the unbound state. This prediction is consistent with the missing satellites problem, where dark matter halos below a critical velocity dispersion $\sigma_{\rm crit} \approx 4\ \mathrm{km\,s^{-1}}$ disperse rather than virialize, because no local black hole horizon is available to absorb the entropy produced at the virialization boundary [Mahaffey, 2026e].

7.2 Landauer's principle as the physical basis for scale invariance

The virial theorem has been verified across 37 orders of magnitude without a physical explanation for why it should hold across such a range. The thermodynamic identity provides that explanation. Landauer's principle applies wherever an irreversible process produces entropy. The virialization boundary exists wherever a $1/r$ potential produces a bound state. The boundary condition $\langle K \rangle = Q$ therefore holds at every such scale, from the hydrogen atom to the cosmic horizon, because Landauer's principle is scale-invariant and the $1/r$ potential form is universal.

This addresses the question of why the virial theorem should apply at cosmological scales. It is not an extension of a mechanical result beyond its domain. It is a thermodynamic identity that applies wherever its conditions are met, at any scale.

7.3 Relationship to Jacobson 1995

Jacobson Jacobson [1995] derived the Einstein equations from the thermodynamic identity $\delta Q = T\,\delta S$ applied to the Rindler horizon of an accelerated observer, with $S = A/4\ell_P^2$. That derivation showed that Einstein's equations are thermodynamic in origin.

The present derivation applies the same thermodynamic identity $Q = T\,\Delta S$ to the virialization boundary of a self-gravitating system and recovers the virial theorem. Both derivations start from the same thermodynamic identity. Both recover a fundamental result of classical physics from thermodynamic first principles. The present work extends the Jacobson program from the derivation of field equations to the derivation of the equilibrium condition governing bound states under those equations.

8 Conclusion

The virial theorem $2\langle K\rangle + \langle V\rangle = 0$ has been verified across 37 orders of magnitude for 154 years without a derivation from thermodynamic first principles and without a physical identification of what K represents beyond the average kinetic energy of bound particles.

This paper shows that $\langle K\rangle$ is the heat released irreversibly at the virialization boundary, encoded as entropy on the nearest available surface at Landauer cost $k_B T \ln 2$ per bit. The chain $\langle K\rangle = Q = T\,\Delta S = E_{\text{Landauer}}$ is exact, derived from the first law, the second law, and Landauer's principle.

The virial theorem is the mechanical expression of this thermodynamic identity. The $1/2$ partition is the unique ratio at which mechanical and thermodynamic equilibrium are simultaneously satisfied for a $1/r$ potential. Its appearance across 37 orders of magnitude is a consequence of the scale-invariance of Landauer's principle and the universality of the $1/r$ potential form.

Three of the four fundamental forces satisfy the identity. The weak nuclear force is the structural exception, excluded by physical necessity. The data are consistent with the identity at 0.2σ at cosmological scales from 17 independent MCMC chains with zero free parameters.

We propose that this thermodynamic identity constitutes a fundamental law of physics of which the virial theorem is the mechanical special case.

Acknowledgments

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Data Availability

All MCMC chains, analysis scripts, and supporting papers are publicly archived at `https://github.com/hmahaffeyges/IAM-Validation` with canonical Zenodo DOI 10.5281/zenodo.18702042.

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